Quantitative Morphometry Handout: Understanding Automated Grain Extraction & Porosity Segmentation

Llinersy Uranga Piña, LPPI - Cergy-Paris Université

June 2026

1 Overview

Microscopic imaging techniques such as Scanning Electron Microscopy (SEM) or Transmission Electron Microscopy (TEM) are fundamental for characterizing the morphology of grains, catalysts, and membranes. Traditionally, researchers analyze these images manually, using cross-line measurements or counting features by hand. This introduces substantial personal bias and limits the statistical count of features.

This reference manual provides the physical, mathematical, and algorithmic reasoning behind the automated pipeline implemented in `Module 4: Image Processing & Morphometry`. By modeling micrographs as numerical matrix intensity distributions, we can process and measure hundreds of particles simultaneously with high statistical precision.

2 The Core Image Transformation Pipeline

2.1 Edge-Preserving Smoothing (Gaussian Filtering)

Raw electron micrographs are prone to high-frequency electronic noise, detector temperature variations, and surface charging artifacts. If an adaptive threshold is directly applied to raw data, this noise creates false boundaries.

- **The Operator:** `filters.gaussian(image, sigma=1)` Gaussian smoothing applies a spatial convolution matrix weighted by a 2D Gaussian bell-curve:

$$G(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{1}$$

- **The Knob (σ):** Lower values ($\sigma \leq 1$) eliminate pixel noise while preserving clean particle interfaces. If raised to $\sigma = 5$, the boundaries of individual grains smear together, preventing accurate downstream segmentation.

2.2 Local Adaptive Thresholding

Standard global thresholding methods (like Otsu's threshold) assume a uniform intensity background across the image. However, SEM images regularly display macro-scale lighting gradients due to sample tilt, uneven electron dispersion, or localized shadows.

- **The Solution:** `filters.threshold_local(smoothed, block_size=51, offset=0.02)`

- **Mathematical Baseline:** Instead of computing a single global limit, a distinct threshold $T(x, y)$ is dynamically generated for every pixel based on the mean intensity of its neighboring grid:

$$T(x, y) = \mu_{\text{local}}(x, y) - \text{offset} \tag{2}$$

  The parameter `block_size` must be an odd integer chosen to be larger than the average diameter of a single particle, allowing it to accurately capture background lighting trends.

2.3 Morphological Post-Cleaning

Threshold processing often leaves behind noise elements or incomplete boundaries.

- **`remove_small_objects`:** Automatically eliminates isolated high-intensity clusters smaller than a target pixel threshold value (e.g., 64 px). This filters out tiny random artifacts before they are registered as particles.
- **`binary_closing`:** Dilates the binary structures slightly before eroding them back. This operation fills micro-pinholes inside grain cores and connects small cracks in boundaries without altering the outer particle profiles.

3 Marker-Controlled Watershed Segmentation

When multiple grains touch, standard contour tracking fails, registering them as a single clustered blob. To split these into independent grains, the script uses a marker-controlled watershed segmentation.

3.1 The Distance Transform

The binary mask is converted into a continuous structural landscape using the Euclidean Distance Transform (EDT):

$$D(x, y) = \min_{(x_b, y_b) \in \text{Background}} \sqrt{(x - x_b)^2 + (y - y_b)^2} \tag{3}$$

For any pixel within a particle, $D(x, y)$ equals the exact geometric distance to the closest background pore channel boundary. The core center of a grain naturally becomes a peak maximum in this calculated distance map.

3.2 Peak Seeding & Topographical Flooding

The algorithm extracts these local coordinate maxima (`peak_local_max`) and treats them as isolated seed markers. The mathematical landscape is then inverted ($-D(x, y)$) and "flooded" with fluid from each marker point. Where adjacent rising wavefronts intersect, the algorithm draws an exact physical dividing line, splitting touching particles cleanly.

4 Extracting Solid-State Metrics

4.1 Fractional Porosity (ϕ_{2D})

Porosity controls mass transport properties in thin films, catalyst supports, and battery membranes. The script quantifies the 2D void space ratio directly from the cleaned binary mask matrix:

$$\phi_{2D} = \frac{\sum \text{Pixels}_{\text{Background}}}{\text{Total Image Footprint Area}} \times 100\% \tag{4}$$

In the binary mask, background void pixels evaluate to 0, allowing their total area to be tracked directly using the command `np.sum(cleaned == 0)`.

4.2 Shape Analysis (Aspect Ratio)

For every isolated particle label, the script evaluates its equivalent ellipse tensor matrix, extracting the major and minor axis lengths. The aspect ratio is quantified as:

$$\text{Aspect Ratio} = \frac{\text{Major Axis Length}}{\text{Minor Axis Length}} \tag{5}$$

An aspect ratio of 1.0 indicates a perfectly isotropic spherical particle. Deviations above 1.0 quantify directional elongation, providing an indirect measure of asymmetric crystal growth or directional mechanical strain fields within the material.